Highlights

- Factorial study decomposes framing into text, structure, role, and acoustic style.
- Narrative structure raises attack success by 27.9 pp (OR 16.65).
- Paired cross-modal safety divergence is model-dependent, not modality-intrinsic.
- MSRF is a knowledge-driven fusion of four signals (225 params; ROC-AUC 0.9784).
- Honest reporting: defense gate G2 fails; fixed thresholds do not transfer.

Cross-Modal Safety Inconsistency in Large Audio-Language Models: Measurement, Mechanism, and Knowledge-Driven Multi-Source Fusion

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Abstract

Large audio-language models (LALMs) are deployed in safety-sensitive decision contexts, yet the safety implications of *adversarial framing* — repackaging a fixed request through narrative, role-play, or acoustic style — remain poorly understood. We treat this as a *knowledge-system* problem and propose a knowledge-driven methodology: a safe decision must fuse heterogeneous knowledge sources — request intent, narrative structure, acoustic style, and scoring uncertainty — that current detectors represent inconsistently or not at all. In a controlled paired factorial experiment ($E_t \times N \times R \times A_s$, 24 cells; 16,200 response cells; three LALMs; two languages), narrative structure (N) raises attack success by 27.9 percentage points (95% CI [25.0, 30.9]; odds ratio 16.65 [14.10, 19.67]). The effect is partially—not wholly—length-mediated and robust across languages, models, and out-of-distribution anchors; the agreement-conditioned dual-judge estimate (38.0 pp) is disclosed as selection-inflated. We further show paired cross-modal safety divergence (PCSD) is model-dependent, so audio vulnerability is not intrinsic to the modality. We then build MSRF, a knowledge-driven multi-source fusion framework that fuses these signal families through calibrated out-of-fold scores in an auditable 225-parameter layer. It reaches 0.9784 ROC-AUC on the deployment frame (0.9455 $\pm$ 0.030 cross-validation), outperforming billion-parameter guards, with boundaries reported honestly (75.9% benign false-positive rate; adaptive-attack decay). Narrative structure exhibits a robust causal effect under controlled prompt interventions; representing that

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effect as structured, multi-signal knowledge is a more durable foundation for LALM safety than either modal shortcuts or opaque detection.

Keywords: cross-modal consistency, knowledge-driven fusion, LALM safety, framing attack, causal attribution

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1. Introduction

Large audio-language models (LALMs) now power voice assistants, multi-modal customer service, and hands-free interaction, and are increasingly deployed in safety-sensitive contexts. A growing body of work shows that such models can be steered toward harmful output by *adversarial framing*: keeping the underlying request fixed while changing how it is packaged through narrative, role-play, or acoustic style (Yu et al., 2026; Qian and Li, 2026; Li et al., 2026). These attacks are practical — they require no gradient access, operate over the channel the model was designed to consume, and are largely invisible to text-only safeguards.

Yet the mechanisms behind framing attacks remain poorly understood. Prior work treats "delivery" as a single coarse preset and compares whole attacks against whole defenses. It does not ask *which semantic component* — the emotional narrative text, the event-chain structure, the role assignment, or the acoustic style — drives attack success, nor whether these components interact. Moreover, causal attribution has remained at the micro (internal) layer where it is inaccessible to black-box practitioners, and existing audio safeguards are rudimentary relative to their text counterparts.

Obstacles to component-level analysis.. Three obstacles have blocked component-level analysis. First, framing components are confounded with response length: narrative prompts elicit longer model outputs, and length itself correlates with harmful content, so a naive comparison cannot separate framing from verbosity (Section 6.1.4). Second, safety judgment is itself unreliable: single classifiers fail catastrophically on Chinese inputs (FNR $\geq$ 0.96 for three off-the-shelf scorers) and disagree with each other on roughly one in six responses, so any measurement must rest on an audited multi-scorer protocol (Section 6.4.4). It also means a reliable safety decision cannot trust any single source and must fuse heterogeneous, mutually non-overlapping signals (Section 5). Third, large audio-language models form a heterogeneous family — a finding that holds for one architecture may not hold for another (Section 6.2).

Connection to Knowledge-Based Systems.. We cast the problem as a *knowledge-system* problem rather than only a safety one. The four signals our study manipulates or measures — request intent, narrative structure, acoustic style, and scoring uncertainty — are four heterogeneous knowledge sources about a single decision (is this interaction harmful?). MSRF is a knowledge-fusion method that represents each source's evidence, calibrates it, and combines it under uncertainty into an auditable score. The paired cross-modal divergence (PCSD) quantifies a *cross-modal knowledge inconsistency* — the same request elicits different safety-relevant knowledge through text versus audio. This aligns the work with the knowledge-systems tradition of heterogeneous knowledge fusion, uncertainty reasoning, and decision support (Guo et al., 2024; Shao et al., 2024; Xu and Li, 2025; Zhao et al., 2025; Shafer, 1976) and with recent KBS contributions that treat safety as a knowledge- representation and decision-support problem (Shi et al., 2026; Yi et al., 2024).

Contributions.. We report a controlled, paired factorial study that decomposes the four framing components across text and audio channels, a response-level cross-modal consistency metric, and a signal-level fusion framework — treating attack mechanisms as structured, auditable, multi-signal knowledge rather than opaque activations, an orientation shared with knowledge-based, decision-support systems (Shi et al., 2026; Yi et al., 2024). Concretely:

(i) **A knowledge-driven multi-source fusion framework.** We build Multi-Source Risk Fusion (MSRF), which represents four mutually non-overlapping signal families (harmful-intent semantics, narrative structure, acoustic prosody, and scoring uncertainty) as explicit knowledge sources and fuses them through out-of-fold calibration and a 225-parameter, auditable fusion layer. On the deployment frame it reaches 0.9784 ROC-AUC (0.9455 ± 0.030 over five frames), outperforming ShieldGemma, WildGuard, and a reproduced GradSafe (ROC-AUC 0.198, 0.174, and 0.315, respectively) on the same test frame, while using roughly two–three orders of magnitude fewer parameters. We report its limits honestly: fixed-threshold detection does not transfer to benign inputs (75.9% benign false-positive rate at the calibrated threshold), and adaptive attacks degrade detection (up to 20.0 pp). We therefore present MSRF as a *discriminative,* *in-distribution* framework for mechanism-aware risk scoring, not as a turnkey deployment defense.

(ii) **Full-factorial causal attribution.** We manipulate E_t (narrative text framing), N (narrative structure / event-chain), R (role assignment), and A_s (acoustic style) in a 24-cell paired design over three large audio-language models, two languages, and an out-of-distribution anchor set. Narrative structure is the dominant causal driver: on the full sample, attack success rises from 2.2% to 30.1% (Δ27.9 pp [25.0, 30.9]; odds ratio 16.65 [14.10, 19.67]), with a consistent effect across languages (zh 29.3 pp / en 26.6 pp), across models, and on the out-of-distribution AdvBench anchor (Δ38.0 pp, dual-judge robustness caliber). A response-length covariate mediates a substantial share of the raw effect, supporting an *elaboration-channel* account rather than a modality-intrinsic one. Role assignment has a *negative* effect (odds ratio 0.53) — a result we replicate and interpret rather than smooth over.

(iii) **Response-level cross-modal consistency.** We quantify paired cross-modal safety divergence (PCSD) as a paired consistency score over the safety-relevant properties of the text and audio rendering of the same input, and show it is model-dependent (e.g., near-zero divergence for one architecture, strongly positive for two others) — evidence that audio "vulnerability" is not intrinsic to the modality. We report it as an auxiliary diagnostic rather than a standalone contribution.

We pre-registered two decision gates — G1 (attribution established) and G2 (defense value); G1 passes, G2 does not, and we report both outcomes rather than only the defense-positive one. The remainder of this paper presents the protocol and factorial design (Section 3), the knowledge-driven fusion framework and its formalization (Section 5), the attribution results (Section 6.1), the cross-modal analysis (Section 6.2), the fusion detection results and their honest boundaries (Section 6.3), adaptive and degradation robustness (Section 6.4), and discussion (Section 7), followed by limitations (Section 8) and conclusion (Section 9). Table 1 positions this work against the most related prior work across the four dimensions we contribute.

Table 1: **Positioning against the most related prior work.** Compared across the four dimensions of this paper's contribution. ✓ = present; ○ = partial (a single framing factor isolated without full decomposition); (✓) = a benchmark-level aggregate rather than a response-level measurement. All attributes follow the characterizations in Section 2.

Work	Component-level attribution	Cross-modal consistency	Fusion-based defense	Honest boundary reporting
Yu et al. (2026)	–	–	–	–
Qian and Li (2026)	○	–	–	–
Li et al. (2026)	–	–	–	–
Pan et al. (2025)	–	(✓)	–	–
Xu et al. (2024)	–	✓	–	–
This work	✓	✓	✓	✓

2. Related Work

2.1. Framing and delivery attacks against LALMs

Audio-side attacks on large audio-language modelshave recently been shown to succeed via narrative delivery (Yu et al., 2026) and prosody (Qian and Li, 2026), and style-aware audio jailbreaks (Li et al., 2026) demonstrate that acoustic style alone can steer output. Paralinguistic variation, such as speaker emotion, similarly induces safety inconsistencies (Feng et al., 2026b), and a recent taxonomy organizes audio jailbreaks into semantic, acoustic, signal, and embedding layers, with narrative framing as a semantic-layer threat (Feng et al., 2026a). These works establish that framing is a real attack surface, but treat delivery as an atomic preset; none decomposes the semantic components or measures their causal contributions under control (Qian and Li, 2026; Yu et al., 2026; Li et al., 2026; Feng et al., 2026b,a). Our design differs on exactly this axis: where Yu et al. (2026) and Li et al. (2026) compare whole attacks against whole defenses, we decompose the "storytelling" bundle into four orthogonal components (text, structure, role, acoustic style) and attribute success to each under a paired control. Where Qian and Li (2026) varies prosody in isolation, our acoustic factor is only one of four, and we disclose that it confounds speaker identity and gender rather than claiming it isolates prosody (Section 4). Text-side research has taxonomized such persuasion strategies (Zeng et al., 2024b); our factorial design is the component-level complement that quantifies their causal contribution in the audio-language setting. The increment over this line of work is the decomposition itself: rather than proposing a new delivery variant, we provide a reusable, controlled measurement of *which* framing component carries the effect, under a paired design that separates structure from text, role, and acoustic style.

2.2. Safety benchmarks and consistency measurement

Benchmarks such as HarmBench (Mazeika et al., 2024), Audio Jailbreak (Song et al., 2025), and Omni-SafetyBench (Pan et al., 2025) evaluate whether models or guards are safe on static collections, and Omni-SafetyBench further defines a benchmark-level cross-modal consistency score (CMSC). Consistency at the benchmark level, however, aggregates outcomes and cannot localize per-input divergence between the text and audio rendering of the *same* query — precisely where deployment disagreements originate. We therefore define PCSD at the response level as a measurement, not a benchmark, and operationalize the distinction in Section 6.2 (Pan et al., 2025). The increment is response-level granularity: PCSD scores the text and audio rendering of the *same* input per cell, which a benchmark-level aggregate cannot localize, and it is reported as an auxiliary diagnostic signal rather than a label.

2.3. Jailbreak detection and defense

Jailbreak behavior in aligned models has been traced to brittle safety-training boundaries (Wei et al., 2023), and detection approaches span gradient analysis (Xie et al., 2024a), input-output safeguards (Inan et al., 2023; Zeng et al., 2024a; Han et al., 2024), cross-modality consistency checking for multimodal LLMs (Xu et al., 2024), confidence- and codebook-based detection (Chen et al., 2025; Alanova et al., 2026), and audio authenticity verification (Xie et al., 2024b). These detectors share a single-cue design: consistency disagreement (Xu et al., 2024) or first-token confidence (Chen et al., 2025; Alanova et al., 2026) alone, with no explicit packaging-structure signal. Our fusion framework instead combines four mutually non-overlapping, interpretable signal families and, unlike prior work, is evaluated on both in-distribution discrimination and honest deployment boundaries (Section 6.3.4). On the defense side, inference-time refusal steering for large audio-language models has also been proposed (Lin et al., 2026). Within the *Knowledge-Based Systems* scope, jailbreak mitigation has been framed as a model-agnostic semantic problem (Shi et al., 2026) and safety as a subspace-oriented model-fusion objective (Yi et al., 2024); the journal's orientation toward knowledge representation and interpretable, decision-support detection is our target (Shi et al., 2026; Yi et al., 2024). Relative to this literature, our contribution is three-fold: (a) causal attribution of *which* framing component matters (a knowledge question, not a classifier question); (b) a fusion framework whose branches are mutually non-overlapping, interpretable signal families rather than opaque embeddings; and (c) an honest evaluation that separates in-distribution discrimination from deployment-level threshold behavior, which most detector papers do not report.

2.4. Knowledge fusion and decision-support systems

Within the knowledge-systems tradition, safety-relevant and operational decisions are supported by fusing heterogeneous knowledge sources under an explicit treatment of uncertainty: belief-theoretic uncertainty reasoning and quantification (Guo et al., 2024), evidential (Dempster–Shafer) fusion of multimodal information for reliable decisions (Shao et al., 2024), multi-criteria fusion over multi-source decision data (Xu and Li, 2025), and ensemble decisions over multimodal decision-information systems under uncertainty (Zhao et al., 2025). These works share a methodology that this paper adopts: represent each source's evidence explicitly, calibrate or discount it, and combine it under uncertainty to support a decision — the same structure that MSRF instantiates for LALM safety scoring with four mutually non-overlapping signal families (Shafer, 1976). Our increment over this line is twofold: the knowledge sources are *response-level signals extracted under a pre-registered causal protocol* rather than generic features, and the fused decision is evaluated both for in-distribution discrimination and for its honest boundary behavior (fixed-threshold transfer), which decision-fusion studies typically do not report.

3. Protocol and Factorial Design

> Full machine-readable protocol: `RESEARCH_PROTOCOL.md` v6.5 and `STAGE_CONTRACTS.md`; this section is the paper-facing summary.

3.1. Research questions and gates

RQ1 (attribution). Which framing component(s) causally increase attack success in large audio-language models, and are effects robust across language, model, and out-of-distribution inputs?

Table 2: **The four framing factors.** Varied in the paired factorial design (§4).

Factor	Definition	Levels
E_t	Narrative text framing (emotional / narrative wording)	neutral vs. narrative
N	Narrative structure (event-chain packaging)	direct request vs. chained narrative
R	Role assignment (identity + task)	none vs. expert role
A_s	Acoustic style (TTS rendering)	neutral voice vs. styled voice

RQ2 (cross-modal). Do text and audio renderings of the same input diverge in safety judgment, and is the divergence model-dependent?

RQ3 (detection). Can a structured, signal-level fusion framework discriminate framed attacks in-distribution, and how does it behave under adaptive attack and audio degradation?

We pre-registered two decision gates. **G1** (attribution established) requires a consistent, ≥10 pp effect whose bootstrap CI excludes zero under the dual-judge caliber with acceptable scoring dispute. **G2** (defense value) requires a significant TPR gain, ≥3 contributing branches, and a benign false-positive rate ≤5%. As reported in Sections 6 and 7, G1 passes and **G2 does not pass**; the paper therefore presents the attribution and mechanism results as primary and the fusion framework as an application-level validation with explicit boundary conditions.

3.2. Factors and manipulation

A paired factorial design (Fisher, 1935) varies four framing components while holding the harmful request fixed:

Three templates instantiate each condition (9 templates total); a manipulation check on response-level markers confirms each textual factor was delivered (mean marker deltas 0.37–0.60, threshold 0.95 keep rate 1.0). We disclose that the acoustic factor is implemented via TTS voices (a female neutral voice vs. a male styled voice), so it confounds speaker identity and gender with prosody. We do not claim A_s isolates prosody, and we verify no systematic length difference across A_s variants (mean difference 1.18 chars, $p = 0.24$).

4. Experimental Setup

4.1. Models and scorers

Three large audio-language models constitute the model matrix: the two primary Gemma-4-family models (E4B-it, audio+text; E2B-it, text) and Qwen2-Audio-7B as an architecture control that is excluded from primary-effect inference. All Gemma-family models run at BF16 with thinking disabled.

Safety judgment uses an audited multi-scorer protocol. Four scorers (HarmBench-cls (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024), and two Gemma-family judges) are validated on an internal validation set (JailBench, $n = 602$)[1]; the primary judge (judge_big) reaches 0.8555 accuracy. We define three calibers: *primary* (single best judge, full sample), *dual-judge* (two judges agree, else abstain), and *majority* (four-vote). We use the **primary** full-sample estimate for headline claims because it is the only caliber free of a selection bias we disclose next. We use the **dual-judge** caliber for cross-scorer robustness and for Chinese-language measurement, where single scorers fail catastrophically (HarmBench FNR = 1.0; StrongREJECT FNR = 0.99) while dual-judge reaches FNR 0.29 with 85.3% accuracy (the third off-the-shelf scorer, the Gemma-family judge, has FNR = 0.96). Because the dual-judge and majority calibers are defined only on rows where the two judges agree, and agreement correlates with condition (in the confirmatory experiment, storytelling rows are included at 57.9% vs. 91.8% for baseline), these calibers systematically inflate the effect estimate (Δ38.0 vs. 27.9 pp). We report this agreement-conditioning selection bias in Section 6. We further disclose that all internal scorers are Gemma-family (three of the four majority votes; the four-vote majority double-counts the E2B family leg), and that agreement with an independent cross-family anchor (Qwen2.5-32B-Instruct-AWQ) is high (0.929 on the E4B side; 0.962 on the full E2B text set), which bounds the same-family risk.

[1] JailBench is not a public benchmark; it is reported here for reproducibility.

4.2. Statistical protocol

Effects are reported as paired per-query cluster bootstraps (Efron and Tibshirani, 1993) (B = 10,000) with 95% CIs, plus a Binomial Bayesian mixed-effects model (Bates et al., 2015) with query random intercepts and fixed model/template effects (the mixed-effect upgrade to full random structure is not met — fewer than three models and three templates — and is reported as such; the benchmark-level cross-frame fusion table uses B = 2,000). Paired per-query discordance is additionally tested with McNemar's test (McNemar, 1947) on the baseline-safe/storytelling-harmful versus baseline-harmful/storytelling-safe cells; the asymmetry is strong across languages and models (Chinese χ^2=34.6, English χ^2=30.3, E2B χ^2=83.2, E4B χ^2=91.1; all $p < 0.001$), and the same test is applied after 1:1 nearest-neighbor length matching (caliper 0.2 SD). We pre-registered TOST equivalence bounds (±5 pp) (Lakens et al., 2018), O'Brien–Fleming sequential alpha spending, and manipulation-check thresholds. All generation is deterministic (byte-identical on re-run, 40/40), and decoding robustness is verified under sampling (Δ20.7 pp greedy vs. Δ20.8 pp sampled, $n = 48$).

5. A Knowledge-Driven Multi-Source Fusion Framework (MSRF)

Single-cue detectors are not trustworthy on framed inputs: off-the-shelf harmfulness classifiers fail catastrophically on Chinese inputs (FNR ≥ 0.96; Section 4) and the strongest open guards score below chance on the framing distribution (Section 6.3). The preceding analysis also establishes that the effect is carried by a specific, identifiable component — narrative structure — not by generic "attackness" or modality. MSRF is a knowledge-driven response to both facts: it represents the evidence about a response as *four mutually non-overlapping knowledge sources*, calibrates each source's score, and fuses them under an explicit uncertainty treatment into a single auditable score. Figure 1 summarizes the architecture.

5.1. Signal families as heterogeneous knowledge sources

Each signal family is a knowledge source $i \in \{\text{int}, \text{nar}, \text{ac}, \text{unc}\}$ that produces a score $x_i(r) \in [0, 1]$ for a response r (its text or audio rendering of the

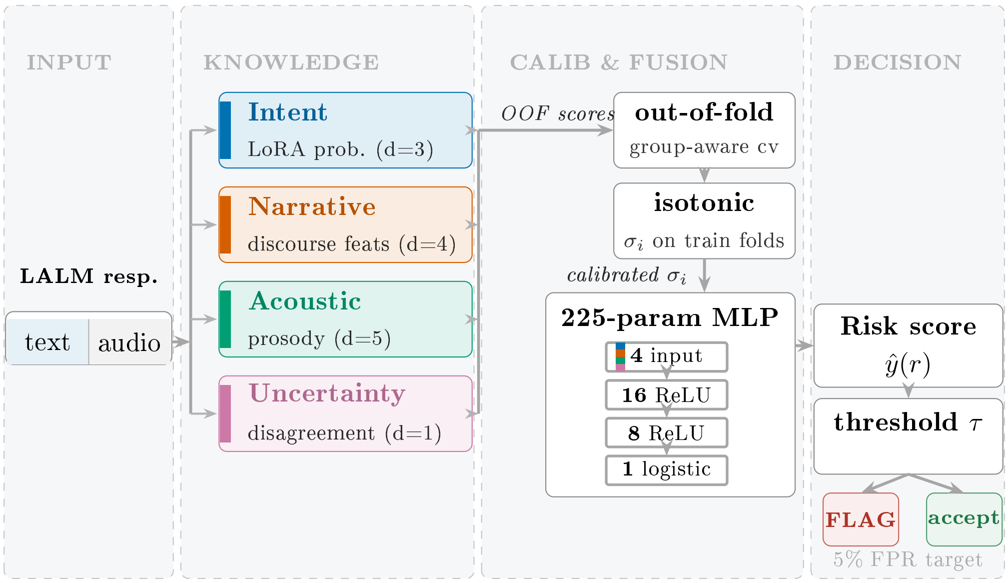

Figure 1: **The MSRF knowledge-driven fusion framework.** Four mutually non-overlapping knowledge sources — harmful-intent semantics (LoRA probability, d=3), narrative structure (discourse features, d=4), acoustic prosody (d=5), and scoring uncertainty (judge disagreement, d=1) — are extracted from each response, scored out-of-fold under group-aware splits (OOF scores), calibrated with isotonic regression (σ_i), and fused by a 225-parameter MLP ($4 \to 16 \to 8 \to 1$) into a discriminative risk score $\hat{y}(r)$; the operating threshold τ targets a 5% FPR operating point on the deployment frame. The honest boundary behavior of the framework (G2) is reported in Section 6.3.4.

same query):

$$x_{\text{int}}(r) = g_{\text{int}}\big(p_{\text{int}}(r),\, p_{\text{int}}(r)^2,\, \ln\big(1 + \ell(r)\big)\big), \tag{1}$$

$$x_{\text{nar}}(r) = g_{\text{nar}}\big(\ell(r),\, n_{\text{sent}}(r),\, d_{\text{nar}}(r),\, n_{\text{story}}(r)\big), \tag{2}$$

$$x_{\text{ac}}(r) = g_{\text{ac}}\big(f_0(r),\, e(r),\, \delta(r),\, \text{zcr}(r),\, r_{\text{sp}}(r)\big), \tag{3}$$

$$x_{\text{unc}}(r) = 2 \cdot \min\big(p_{\text{unc}}(r),\, 1 - p_{\text{unc}}(r)\big). \tag{4}$$

The intent source x_{int} is a LoRA fine-tuned harmful-intent classifier over the E2B base whose input features are the intent probability, its square, and normalized response length ((1)); the narrative source x_{nar} uses interpretable discourse-structure features — length, sentence count, narrative density, and story-word presence ((2)); the acoustic source x_{ac} uses prosodic features (f0, energy, duration, zero-crossing rate, speech/pause ratios) with mean-imputation-plus-mask when audio is missing ((3)); and the uncertainty source x_{unc} encodes scoring disagreement across the multi-scorer protocol as $2 \cdot \min(p, 1-p)$, maximal when the scorer consensus is maximally uncertain ((4)). The families are designed to be non-overlapping: each covers a disjoint aspect of the response, so the fusion cannot "double count" the same evidence through two branches. We disclose two boundary conditions of the sources: the intent branch was trained in-distribution (its training rows overlap the evaluation set by design of the stacking protocol), and the acoustic branch has no measurable signal at our sample window ($N = 32$ attacks). We phrase the latter as "not measured", not "ineffective".

5.2. Fusion rule and parameter budget

Branch scores are calibrated with isotonic regression σ_i fit on training splits to render them comparable across families, then fused by a two-hidden-layer MLP $\mathcal{F}$:

$$\hat{y}(r) = \mathcal{F}\Big(\sigma_1\big(x_{\text{int}}(r)\big),\, \sigma_2\big(x_{\text{nar}}(r)\big),\, \sigma_3\big(x_{\text{ac}}(r)\big),\, \sigma_4\big(x_{\text{unc}}(r)\big)\Big), \tag{5}$$

$$|\mathcal{F}| = (4 \cdot 16 + 16) + (16 \cdot 8 + 8) + (8 \cdot 1 + 1) = 225. \tag{6}$$

The fused layer is a $4 \to 16 \to 8 \to 1$ ReLU MLP with 225 trainable parameters ((6)); the operating threshold τ targets 5% FPR on the deployment frame. The parameter budget is a deliberate design choice: with 225 parameters the entire decision function is inspecting three numbers per source, so every contribution to the score can be traced back to an interpretable branch. This is an auditability property that multi-billion-parameter guards (ShieldGemma-9B, WildGuard, GradSafe) do not offer, and one we exploit to localize the honest boundary of the framework in Section 6.3.4. This is the knowledge-fusion position of Shafer (1976) and recent evidential fusion work (Guo et al., 2024; Shao et al., 2024; Xu and Li, 2025): evidence is made explicit and combined under uncertainty, rather than buried in a black box.

5.3. Evaluation protocol

Branch scores are produced *out-of-fold* so that no row contributes its own label to the score the fusion layer sees. The split is a group-aware GroupShuffleSplit whose group key is template family $\times$ attack family (21 groups), so that an entire template$\times$attack group is held out together and leakage across conditions is prevented. An outer 25% test frame never participates in hyperparameter selection, and an inner 20% validation split selects the MLP learning rate. The protocol is repeated over five independent seeds,

Table 3: **The 225-parameter fusion budget.** Layer-by-layer composition of the fusion MLP $\mathcal{F}$ ($4 \to 16 \to 8 \to 1$). The four inputs are the isotonic-calibrated branch scores $\sigma_1, \ldots, \sigma_4$ (intent, narrative, acoustic, uncertainty); hidden activations are ReLU and the output is a logistic layer, matching the scikit-learn `MLPClassifier` used in the protocol (`hidden_layer_sizes=(16,8)`, `activation="relu"`).

Layer	Shape	Composition	Parameters
1	$4 \to 16$ (ReLU)	$4 \times 16 + 16$	80
2	$16 \to 8$ (ReLU)	$16 \times 8 + 8$	136
3	$8 \to 1$ (logistic)	$8 \times 1 + 1$	9
		Total trainable	**225**

yielding five independent test frames with the deployment frame reported as primary. The silver ground-truth label is an agreement-based subset of the multi-scorer protocol (66.2% coverage; mean pairwise agreement 0.8122), used only to train and score the fused detector.

Algorithm 1 summarizes training and inference.

6. Results

Per the pre-registered protocol (Section 3), this section reports the confirmatory attribution test under the primary caliber first, then exploratory decompositions and diagnostic analyses; primary conclusions are anchored to the pre-registered main-effect hypotheses, and all other comparisons are reported as exploratory.

6.1. Attribution (RQ1)

6.1.1. Narrative structure is the dominant causal driver

In the confirmatory full experiment (P1-FULL; 16,200 cells = 600 queries × 3 conditions × 3 templates × 3 models, where the 600 queries are 400 in-distribution — 200 Chinese and 200 English — plus 200 out-of-distribution AdvBench anchors; text modality), storytelling framing raises attack success from **2.17%** to **30.08%** under the primary (judge_big) caliber on the full sample: **Δ27.9 pp** (95% CI [25.04, 30.92]; mixed-model **OR 16.65** [14.10, 19.67] with query random effects). The effect is robust in direction and significance across all three calibers; the consistency calibers yield larger magnitudes (dual-judge Δ38.0 pp [33.98, 42.2]; majority Δ36.8 pp) because they are defined only on rows where the two judges agree, and agreement correlates

Algorithm 1 MSRF training and inference

Require: responses R (text/audio renderings); silver labels $L \subset R$; groups G (template family $\times$ attack family)
Ensure: calibrated fusion model $\mathcal{F}$; threshold τ

procedure TRAIN(R, L, G)
    for each family $i \in \{\text{int}, \text{nar}, \text{ac}, \text{unc}\}$ **do**
        extract features and compute branch score $x_i(r)$ for all $r \in R$
        recompute branch scores out-of-fold over group-aware splits of G
    end for
    fit isotonic calibrators σ_i on the training splits
    select MLP learning rate on the inner 20% validation split
    fit $\mathcal{F} : \mathbb{R}^4 \to [0, 1]$ ($4{\to}16{\to}8{\to}1$) on calibrated branch scores
    set τ to target 5% FPR on the deployment-frame validation set
    return $\mathcal{F}$, $\{\sigma_i\}$, τ
end procedure
procedure INFER(r)
    compute $\{x_i(r)\}$; **return** $\mathcal{F}\big(\sigma_1(x_1), \ldots, \sigma_4(x_4)\big)$
end procedure

with condition (storytelling inclusion 57.9% vs. baseline 91.8%), which systematically inflates these estimates. We disclose this agreement-conditioning selection bias rather than smoothing it. The unrestricted-creativity condition produces an intermediate effect (Δ18.5 pp on the dual robustness caliber), and the out-of-distribution AdvBench anchor reproduces the storytelling effect under the same dual robustness caliber (Δ38.0 pp, computed on paired dual-judge-eligible cells versus 30.8 pp pooled over the full sample; its odds ratio of 363 is inflated by the near-zero baseline and is reported alongside the percentage-point effect).

6.1.2. Cross-lingual and cross-model robustness

The effect is consistent across languages (Chinese Δ29.3 pp, English Δ26.6 pp; interaction OR 1.41, $p < 0.001$, i.e., a modestly stronger effect in Chinese) and across the two primary models (E4B Δ30.8 pp, E2B Δ25.1 pp) under the primary caliber; these per-language and per-model estimates average to $\approx$27.9 pp, consistent with the pooled headline. Under the dual-judge robustness caliber, the benchmark main table reproduces the same ordering across the model$\times$language grid (range 29.9–40.2 pp), and the

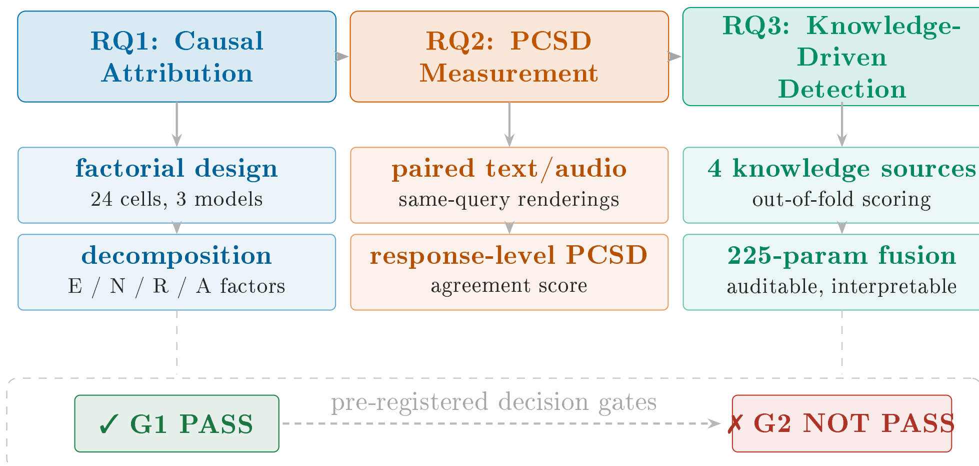

Figure 2: **Study overview: from causal attribution through cross-modal consistency to knowledge-driven detection.** RQ1 attributes attack success to narrative structure (Δ27.9 pp, OR 16.65); RQ2 quantifies response-level paired cross-modal divergence (PCSD) as model-dependent; RQ3 evaluates the knowledge-driven fusion detector MSRF (ROC-AUC 0.9784 on the deployment frame). The pre-registered gates are reported honestly: G1 passes and G2 does not (75.9% benign false-positive rate at the calibrated threshold; Section 6.3.4).

architecture-control model Qwen2-Audio-7B reproduces the direction on text (ASR 5.3%$\rightarrow$38.8%, Δ33.5 pp) while being excluded from the primary mixed model by design. Under our pre-registered cross-lingual consistency rule the effect is judged robust in direction and magnitude.

6.1.3. Benign control: framing amplifies existing intent

On an independent benign-input control ($n = 162$), storytelling framing produces essentially no harmful output (0.0% baseline, 3.2% storytelling, 0.0% unrestricted; both flip directions zero). This supports an *amplification* account: framing does not conjure harmful behavior from benign inputs; it amplifies existing harmful intent.

6.1.4. The elaboration-channel hypothesis: length mediates a large share

The raw framing effect is confounded with response length: storytelling responses average 937 characters versus 417 for baseline. In the pilot factorial ($n =$ 14,400 cells), including a response-length covariate collapses the N effect in the mixed-effects model (OR 1.49 without length vs. 1.03 with length). On the full data, however, within-length-band analyses retain a substantive

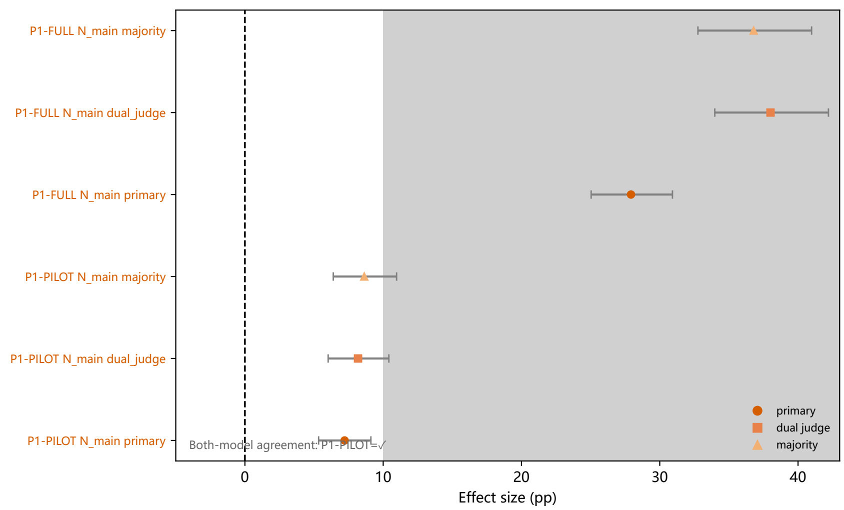

Figure 3: **Factorial forest plot.** Main and interaction effects of the framing components on attack success under the primary caliber, estimated from the pre-registered analysis; the zero line marks no effect, and the shaded band marks the practically meaningful region (≥ 10 pp). Intervals are cluster-bootstrap 95% CIs ($B = 10{,}000$); per-effect n as labeled.

content effect: in the only band with real baseline samples (200–799 chars) storytelling is not separable ($\Delta \approx +0.5$ pp), but in the $\geq$800-char band — where 96% of storytelling responses fall — the storytelling effect is **+21.7 pp** (baseline 9.6% vs. storytelling 31.3%), and length-matching at $\geq$1012 chars still leaves **+17.5 pp**; the judge is not a "long $\Rightarrow$ harmful" shortcut, since baseline long responses score only 9.6% harmful. We therefore characterize the effect as *partially—not wholly—length-mediated* (elaboration-channel: framing $\rightarrow$ longer elaboration $\rightarrow$ more harmful), with a real content effect surviving within matched length bands. On the audio channel the three estimators disagree in sign (single-slope OR 5.20; Mantel–Haenszel OR 0.60; 1:1 matched -10.5 pp, n.s.), and length-matched comparisons render Qwen2-Audio's audio-channel effect negative (-39.3 pp, $p = 0.003$). We report all three estimators rather than a single favorable one, and we do **not** claim that the audio modality is intrinsically more vulnerable.

6.1.5. Component decomposition: N drives, R protects

The factorial decomposition separates the components. Narrative structure (N) carries the effect; narrative text framing (E_t) contributes a smaller but significant effect; **role assignment (R) has a negative effect** (OR

0.53, replicated across models) — assigning the model an expert identity makes harmful output *less* likely in our protocol, a counter-intuitive result we replicate and discuss in Section 7. Interactions are modest: $E_t \times R$ is significant in the pilot (OR 2.54), and the N effect is stronger on audio than on text (pilot dual-judge diff 7.26 pp; majority 9.84→22.66 on audio vs. 3.7→8.38 on text). Family-level stratification shows the effect generalizes across all five covered attack families (5/5 positive, 3/5 significant on dual-judge), with hate speech the most amplified ($\Delta \approx +17.6$ pp, roughly three times the overall effect; $N\times$hate interaction significant in both models, +13.5 pp on E2B and +31.2 pp on E4B); coverage excludes network attacks and benign requests. Of 84 pre-registered comparisons, 65 are significant, all surviving BH-FDR correction (Benjamini and Hochberg, 1995) at $q = 0.10$.

6.2. Cross-modal consistency (RQ2)

We quantify **paired cross-modal safety divergence** (PCSD) through a per-input *agreement* score over the safety-relevant properties of a model's response to the text rendering versus the audio rendering of the *same* query (length ratio and lexical overlap, combined 0.5/0.5; 0–1, higher agreement = less divergence). PCSD is a response-level measurement, distinct from benchmark-level consistency scores (Pan et al., 2025), and we use the complementary audio-minus-text attack-success gap Δ as a per-model divergence index.

Across the model matrix, PCSD is **model-dependent**: agreement is lowest for E4B (0.248) and highest for Qwen2-Audio (0.404), and the audio-minus-text gap mirrors this ordering (E4B +30.7 pp, E2B +44.6 pp, **Qwen2-Audio −0.2 pp**). Two implications follow. First, audio "vulnerability" is not intrinsic to the modality — one architecture shows no audio-text divergence at all. Input-side paralinguistic variation, such as speaker emotion, likewise shifts large audio-language model safety outcomes (Feng et al., 2026b), reinforcing that inconsistency is a property of the model–input interaction rather than of audio itself. Second, cross-modal safety inconsistency is a measurable, per-input phenomenon that a single modality-level metric would obscure; this motivates treating cross-modal divergence as an auxiliary diagnostic signal rather than a label.

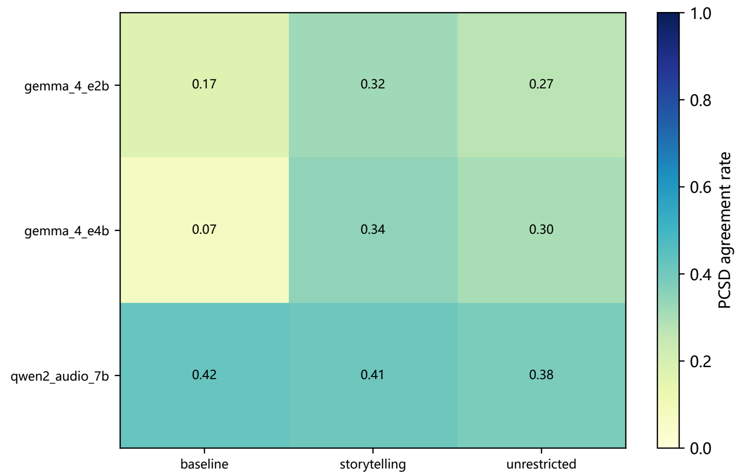

Figure 4: **PCSD paired-divergence heatmap.** Response-level agreement and divergence asymmetry between the text and audio rendering of the same query under attack conditions, across models and framing conditions; darker cells indicate higher agreement (less divergence). Per model $\times$ condition cell n as labeled.

6.3. A knowledge-driven fusion framework (RQ3)

6.3.1. Design

MSRF is presented as a self-contained method in Section 5: four mutually non-overlapping knowledge sources (harmful-intent semantics, narrative structure, acoustic prosody, scoring uncertainty), out-of-fold group-aware scoring, isotonic calibration, and a 225-parameter auditable fusion layer. We restate the headline protocol here for continuity and evaluate the framework below.

6.3.2. In-distribution discrimination

On the deployment frame (n = 3,863, 3.6% positive), MSRF reaches **ROC-AUC 0.9784**, PR-AUC 0.6119, TPR@FPR5 0.8345, ECE 0.0299. Over five independent group-split frames (protocol in Section 5; each frame is that seed's group-aware test split, with the MLP learning rate selected on an inner 20% validation split so test rows never participate in selection), mean ROC-AUC is **0.9455 ± 0.030** (min 0.8909, max 0.9784). The weakest frame still outperforms the strongest open-source classifier by 2.8× (Table 4).

Leave-one-branch-out ablation, evaluated relative to the five-frame mean AUC (0.9455), shows intent and narrative carry the signal (removing them

Table 4: **Cross-frame evaluation of the fused detector.** Five independent group-aware GroupShuffleSplit test frames (group key = template family × attack family, 21 groups; outer 25% test, inner 20% validation for learning-rate selection; Section 5). Frame 20260811 is the deployment frame used for all baseline comparisons.

Seed	Test n	ROC-AUC	PR-AUC	TPR@FPR5
20260811 (deployment)	3,863	**0.9784**	0.6119	0.8345
20260812	2,198	0.8909	0.6804	0.4608
20260813	3,323	0.9447	0.4750	0.6071
20260814	4,242	0.9667	0.7015	0.7891
20260815	3,149	0.9470	0.5555	0.5539
Mean over 5 frames	–	0.9455 ± 0.030	–	–

Table 5: **Leave-one-branch-out (LOO) ablation and single-branch performance.** Evaluated relative to the five-frame mean AUC (0.9455). The acoustic branch has no measurable signal at our sample window ($N = 32$ attacks; reported as "not measured"), and the uncertainty branch has no standalone score.

Branches	LOO ROC-AUC	Δ vs. full	Single-branch ROC-AUC
Full (all four)	0.9455	–	–
− intent	0.9136	−0.032	0.9325 ± 0.033
− narrative	0.9325	−0.013	0.9136 ± 0.048
− acoustic	0.9449	−0.001	0.5 ± 0.0 (no signal)
− uncertainty	0.9448	−0.001	–

drops AUC to 0.9136 and 0.9325, i.e., by 0.032 and 0.013) while the acoustic and uncertainty branches contribute marginally (removal changes AUC by ≤0.001, to 0.9449 and 0.9448). Table 5 reports the full ablation alongside each branch's standalone score: the acoustic branch alone is chance-level (0.5, no signal at our sample window), and the two best branches alone (intent 0.9325, narrative 0.9136) each fall short of the fused detector, so the fusion gain is not attributable to any single knowledge source.

6.3.3. Comparison with open-source guards

On the same deployment frame, MSRF outperforms ShieldGemma-9B (0.198 ROC-AUC), WildGuard (0.174), and a faithfully reproduced GradSafe (0.315) while using 225 parameters versus multi-billion-parameter guards; measured latency is 0.20 s / 6.7 GB for ShieldGemma and 1.04 s / 4.8 GB

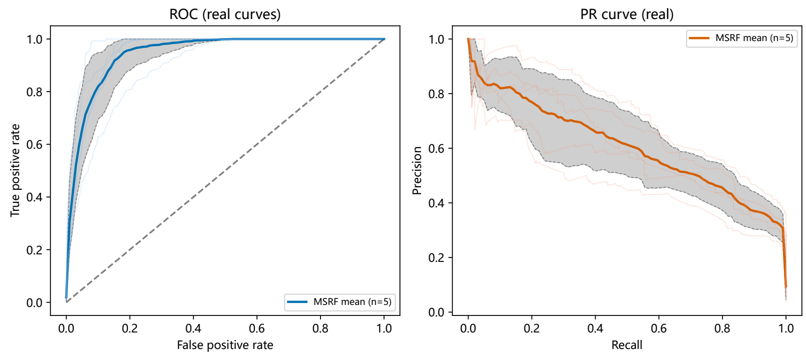

Figure 5: **MSRF ROC/PR curves.** Detection performance of the real fused detector on the deployment frame (n = 3,863, 3.6% positive). Solid curves are the 5-seed mean with a 10–90 percentile envelope (mean ROC-AUC 0.9455 $\pm$ 0.030; the deployment frame achieves 0.9784).

for WildGuard (Table 6). We note that all three open classifiers are *negatively* correlated with our safety labels on framed inputs (ROC-AUC $<$ 0.5), which we interpret as systematic miss-classification of framing (flagging benign, missing harmful) rather than label inversion. Cross-frame replication (five independent frames) confirms MSRF wins every frame against the strongest baseline (margins 0.013–0.037).

6.3.4. Honest boundary: threshold behavior does not transfer

On an independent benign set (n = 162), the calibrated threshold produces a **75.9% false-positive rate** for MSRF, versus 0.0% for ShieldGemma and WildGuard. The root cause is a distribution mismatch: the calibration pool consists of refusal responses, whereas true benign inputs yield compliant responses that score high under intent/narrative signals. We therefore present MSRF as a *discriminative, in-distribution* scoring framework — not a deployment threshold detector — and we report the threshold boundary explicitly. This is the criterion on which our pre-registered defense gate G2 does not pass.

6.4. Robustness and adaptive attack

6.4.1. Defense decay on the attack set

On a separate attack set (n = 2,864, gray-box and white-box perturbations, each class $n \geq 200$), we re-ran the guards at the response level. On the

Table 6: **Open-source guard baselines.** Evaluated on the same deployment frame as MSRF ($n = 3{,}863$). ShieldGemma and WildGuard are real-inference runs; the GradSafe-real row is a reproduction of the ACL 2024 gradient-analysis method; the GradSafe proxy is a zero-additional-model gradient proxy in the same feature space. ROC-AUC below 0.5 indicates negative correlation with our safety labels on framed inputs (systematic miss-classification: flagging benign, missing harmful), which we interpret rather than as label inversion.

Method	ROC-AUC	PR-AUC	Latency	Peak VRAM
MSRF (fusion, 225 params)	**0.9784**	0.6119	–	–
GradSafe proxy	0.9652	0.5961	–	–
ShieldGemma-9B	0.198	0.029	0.20 s	6.7 GB
WildGuard (Mistral-7B)	0.174	0.029	1.04 s	4.8 GB
GradSafe-real	0.315	0.024	–	–

harmful subset ($n = 52$), MSRF detects 15.4%, ShieldGemma 26.9%, WildGuard 48.1%; on all rows, MSRF 19.1%, ShieldGemma 11.3%, WildGuard 22.2%. We report these numbers in full: MSRF's advantage is in-distribution discrimination (AUC), not fixed-threshold hit rate on attacks. We also disclose that the attack responses carry no LoRA intent signal (the intent branch sees an out-of-distribution zero), which explains part of the decay.

6.4.2. Adaptive attack decay

Against the real fused detector, gray-box and white-box perturbation classes show detection-rate decay up to 20.0 pp relative to baseline. We scope the claim precisely: our perturbations are single-pass, prompt-level transformations, not iterative or gradient-based optimization, so we claim *prompt-level perturbation robustness*, not adversarial robustness in the gradient sense. Three of six classes are effectively no-ops (fewer than half the perturbations change the input) and are reported as such rather than as attacks.

6.4.3. Audio degradation

Real audio degradation (Opus 16 kbps, SNR 20 dB, RIR reverberation, each EBU R128-normalized) changes attack success in opposite directions across models (e.g., −7.0 pp for E2B, +11.3 pp for E4B under Opus), which we report as-is. Degraded audio increases "thinking-trace leakage" in one model (10.0% / 26.0% / 28.5% for the three degradations vs. 7.3% baseline), and that model's audio-side robustness must not be extrapolated to its text

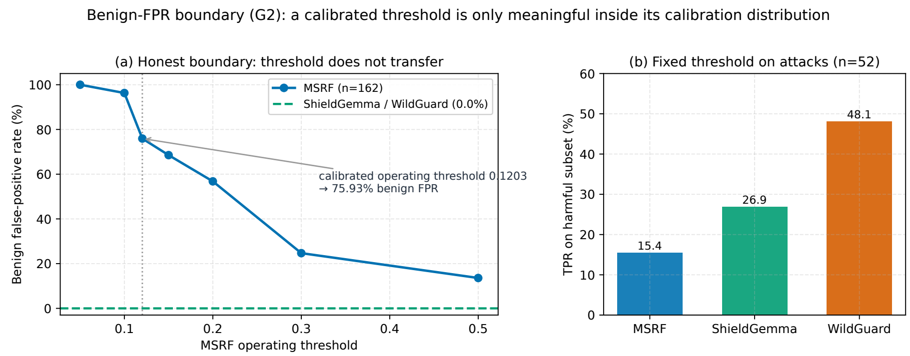

Figure 6: **Benign-FPR boundary (the G2 criterion).** Rendered as evidence rather than avoided: (a) MSRF false-positive rate on the independent benign set ($n = 162$) rises to **75.9%** at the calibrated operating threshold 0.1203, versus 0.0% for ShieldGemma and WildGuard; (b) at that *fixed* threshold, TPR on the harmful subset ($n = 52$) is 15.4% for MSRF, 26.9% for ShieldGemma, and 48.1% for WildGuard — MSRF's advantage is in-distribution discrimination (AUC), not fixed-threshold hit rate on attacks.

side, where empty refusals reach 58–85% (a genuine model behavior we report as such).

6.4.4. Measurement trust chain

We report a five-layer trust chain: deterministic generation (byte-identical, 40/40); scorer reliability (judge_big vs. judge_small agreement 0.82 on n=3,573 aligned cells, with Gwet's AC1 0.745 and balanced κ 0.706 from the pipeline's agreement utilities (Cohen, 1968)); cross-family scorer agreement (E4B side 0.929; full E2B text set 0.962); test–retest agreement (1.000 across three scorers); and Dawid–Skene latent-label recovery (0.945 in simulation at the measured error regime, 50 replications, 180 items, P(harm)=0.5). Batch-inference byte-equality is not guaranteed (42/200 cells byte-identical, zero errors), which we disclose in the reproducibility statement.

7. Discussion

Why narrative structure, and why does role assignment protect?. The factorial results separate the "storytelling" bundle into a strong structure effect (N) and a counter-intuitive protective effect (R). A post-hoc account consistent with both is the elaboration-channel hypothesis: N increases elaboration (length, narrative density), which increases both the probability and

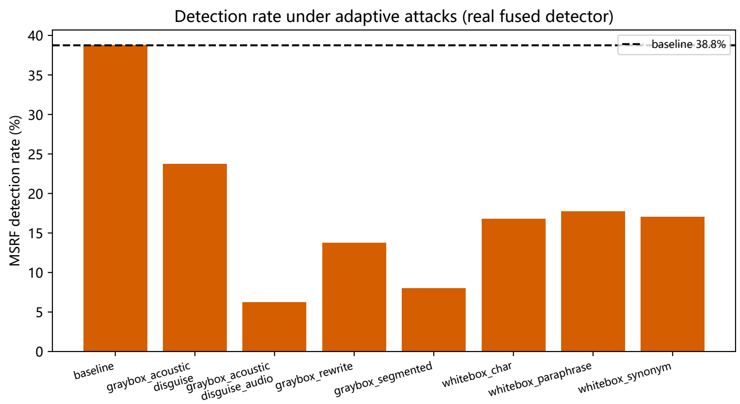

Figure 7: **Detection-rate decay under adaptive perturbations.** The real fused detector under gray-box and white-box adaptive perturbation classes, with decay up to 20.0 pp relative to baseline. Perturbations are single-pass, prompt-level transformations; we claim prompt-level perturbation robustness, not gradient adversarial robustness.

the perceived legitimacy of harmful continuations; R, by contrast, constrains the model into an identity whose stance (an "expert advisor") is at odds with producing harmful instructions in our protocol. The replication of R's sign across models argues against a fluke, but its sensitivity to the scoring caliber (it reverses under latent-label scoring) demands caution; we report both.

Why do open-source guards fail on framing?. ShieldGemma, WildGuard, and GradSafe all score below chance on framed inputs. This is consistent with their training objective (harmfulness of direct requests) being systematically mismatched with framing, which is safe-looking. It motivates a *knowledge* framing of the detection problem: the relevant signal is not "is this text harmful" but "does this structure package harmful intent," which is exactly the signal our narrative and intent branches measure.

Threshold detection versus mechanism-aware scoring.. The benign-FPR failure (Section 6.3.4) is instructive rather than incidental. It shows that a calibrated threshold is only meaningful inside the calibration distribution, and that compliant benign behavior is statistically confounded with framing signals. Read as a *knowledge boundary* rather than a defect, this is a substantive empirical finding about the framework's applicability domain: the calibration pool (refusal responses) and the benign pool (compliant responses) occupy different regions of the fused score space, so the two distributions are not mutually calibrating. It is exactly this kind of domain knowledge that our pre-registered defense gate G2 was designed to elicit, and we report it in full. We argue that the honest contribution of a fusion framework in this setting is a *discriminative risk score with auditable branch attributions* — not a turnkey threshold — and we present the paper's detection claims accordingly. A direct path forward is adaptive or online recalibration on deployment traffic, which we leave to future work. This aligns with the journal's orientation toward interpretable, decision-support knowledge systems (Shi et al., 2026; Yi et al., 2024).

Generalizability of the methodology.. The contribution is best read as a *methodology* rather than a tuned detector. The four knowledge sources — intent semantics, narrative structure, acoustic prosody, and scoring uncertainty — are defined from first principles about the response and the scoring process, not about a specific model or language; the causal protocol and the paired cross-modal measurement are likewise model-agnostic. We therefore expect the framework to transfer to other multimodal model audits (e.g., vision-language assistants whose safety responses also admit packaging structure), to other high-resource languages, and to high-stakes voice applications where an auditable, single-model risk score is preferable to a stack of opaque classifiers. What transfers is the *structure*: mutually non-overlapping, auditable knowledge sources, out-of-fold scoring, and honest boundary reporting. We have not re-run the full protocol on these settings, and we state this explicitly rather than claiming measured transfer.

Implications for knowledge-based decision systems.. For the decision-support community, three design lessons follow. First, a safety decision over a model response is a multi-source fusion problem. The four signals we fuse are heterogeneous knowledge sources with different reliability (the intent and narrative branches carry the signal; the acoustic branch is "not measured", not "ineffective"). Explicit calibration plus an auditable fusion layer lets a system operator trace which source drove a score. Second, the boundary between in-distribution discrimination and threshold transfer is itself decision-relevant knowledge. The calibration pool (adaptive refusals) and the benign pool (compliant responses) live in different regions of the fused score space. A decision system must either keep its operating context inside the calibrated region or re-calibrate online when the deployment distribution drifts toward compliant benign traffic. Third, honest boundary reporting is a decision-support feature, not a weakness: a score with a documented failure region supports risk-informed routing (e.g., escalating scores that fall outside the calibrated envelope to a human reviewer) better than a threshold with an unmeasured false-positive rate. These are the operational postures that follow from our measurement (Guo et al., 2024; Shafer, 1976; Xu and Li, 2025).

8. Limitations

We report the principal limitations explicitly, each with the direction we consider most promising for addressing it.

(i) **Scorer family bias.** All internal scorers are Gemma-family; cross-family anchoring bounds but does not eliminate this. Re-anchoring the protocol with at least one non-Gemma internal scorer to quantify the residual family bias directly is a natural next step.

(ii) **Chinese-language scoring.** Only the dual-judge caliber is reliable on Chinese inputs; single-scorer Chinese results are not reported. Extending single-scorer Chinese reliability (e.g., Chinese-native judges or scorer fine-tuning) is future work.

(iii) **Acoustic factor confound.** A_s is implemented by voice, confounding speaker identity and gender; acoustic-attack signal is not measured at our sample window ($N = 32$). Future work will separate voice identity from prosody (e.g., TTS with matched speaker identity) and enlarge the acoustic window well beyond $N = 32$.

(iv) **Length confound.** The effect is partially—not wholly—length-mediated; within matched length bands a content effect remains, and all three audio-channel estimators are reported. Additional matched-length designs and per-band estimators remain a planned extension.

(v) **Detection transfer.** Fixed-threshold detection fails on benign inputs (75.9% FPR) and decays under adaptive attacks; only in-distribution discrimination and prompt-level robustness are claimed. Adaptive or online recalibration on deployment traffic, and an evaluation under gradient-based adversarial perturbations, are the natural next steps toward threshold transfer.

(vi) **Out-of-distribution intent.** Attack responses lack the LoRA intent signal. Future work will add a LoRA intent signal at inference time or an explicit out-of-distribution guard on the intent branch.

(vii) **Baseline coverage.** JailGuard, Cross-modal Information Check, and SALMONN-Guard were not reproducible (unreleased code/weights) and are reported as cited values only, never as measured baselines. We will re-run these baselines on release of their code or weights.

(viii) **Sequential design.** The pre-registered maximum of 500 cells/condition was exceeded by design change (v6.2→v6.5), which we disclose. A cleaner confirmatory design within the original cap is future work.

(ix) **Missing scorings.** 1.03% of P0-C responses have deterministic parsing failures, bounded and disclosed. Re-scoring those cells with a robust parser is a planned cleanup.

No fabricated values appear anywhere in this paper; every number traces to the released pipeline artifacts.

9. Conclusion

We decomposed adversarial framing against large audio-language models into its semantic components and showed, under an audited paired factorial protocol, that narrative structure robustly increases attack success (Δ27.9 pp, OR 16.65; a larger agreement-conditioned dual-judge estimate is disclosed as inflated by selection) across languages, models, and out-of-distribution inputs. A substantial share of this effect is mediated by response elaboration, and cross-modal safety divergence is model-dependent rather than modality-intrinsic. We further introduced MSRF, a 225-parameter knowledge-driven fusion framework that discriminates framed attacks in-distribution (0.9784 ROC-AUC on the deployment frame) while exposing its own boundary conditions honestly. Narrative structure exhibits a robust causal effect under controlled prompt interventions; representing that effect as structured, multi-signal knowledge is, we argue, a more durable foundation for large audio-language model safety than either modal shortcuts or opaque detection.

Data availability

All pipeline artifacts, configurations, frozen seeds, and result files are released in the public companion repository accompanying this manuscript; no pre-trained guard weights are redistributed.

Ethics

All experiments use fabricated, synthetic harmful queries (never real personal data), are confined to controlled research environments, and are reported under a dual-use disclosure consistent with the study protocol and its gate decisions (see the companion repository). The intent-classifier training data is synthetic.

CRediT authorship contribution statement

Yongjin Chen: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.
Songze Zhu: Funding acquisition.

Declaration of competing interest

None declared.

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Generative AI disclosure

This manuscript was drafted with the assistance of AI writing tools for language and organization; all experiments, measurements, and reported numbers are human-executed and machine-verified.

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